Practice workbook · Say what an algorithm (or one line) does

Reading a supplied flowchart, pseudocode algorithm, subprogram or opened program and explaining in words either its overall job or the role of a particular line or variable (a temporary swap variable, an index increment, a redundant test, a count-down loop variable).

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-02

1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

5 Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

Figure 7

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

(2)

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

(2)

DO NOT WRITE IN THIS AREA
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1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

(b) Martha is writing a program to calculate grades.

Each student takes three tests. Each test has a maximum score of 50.

Scores are stored in a two-dimensional array called tbl_scores.

The lowest of the three scores is ignored and the other two are added to get a total out of 100.

Grades are based on the table in **Figure 8**.

Total	Grade
80 or more	Distinction
65 to 79	Merit
50 to 64	Pass
49 or less	Fail

Figure 8

Each student's total and grade are saved to a text file.

For example, the first student scores 45, 42 and 43. The lowest score 42 is ignored. The total is 88 and the grade is Distinction.

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1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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2. Question 5 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

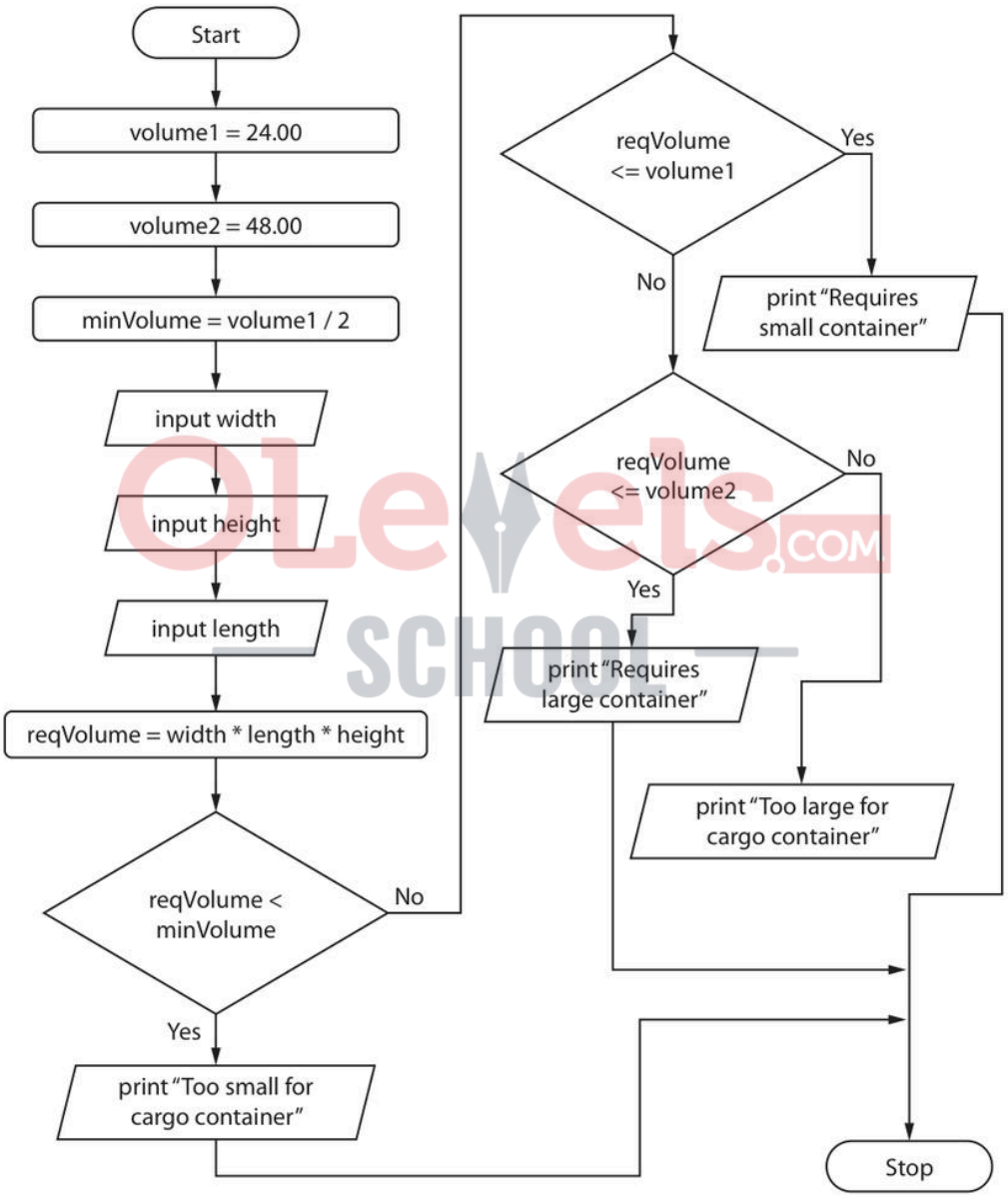


Figure 2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 1 4 2 4

2. Question 5 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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2. Question 5 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

(b) Each ship is registered to a state.

Figure 3 shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO



END FOREACH
```

Figure 3

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 1 6 2 4

2. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

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(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

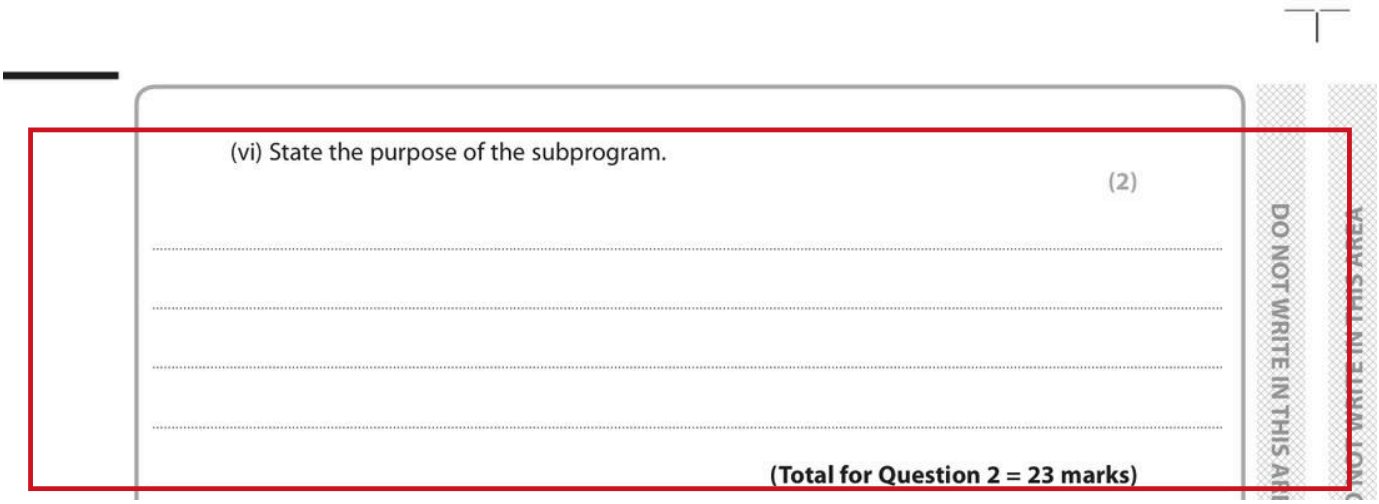
(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

3. Question 2 · part c.vi · 2 marks

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Further practice · Attempt part c.vi only.



(vi) State the purpose of the subprogram.

(2)

(Total for Question 2 = 23 marks)

DO NOT WRITE IN THIS AREA

Working and answer

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4. Question 5 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

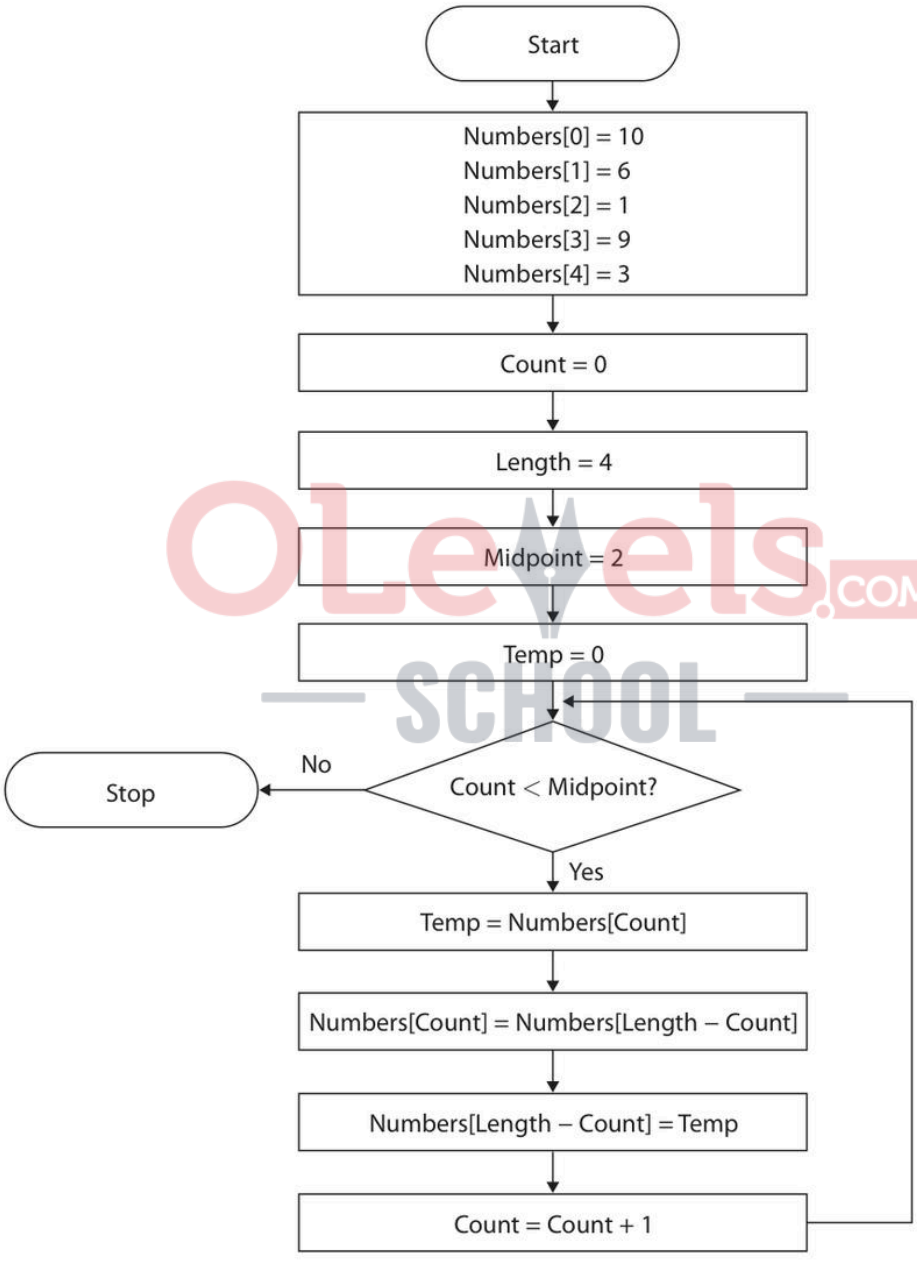


Figure 3

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 2 2 0

4. Question 5 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

.....

.....

(iii) Explain why the variable Temp is needed.

(2)

.....

.....

.....

.....

4. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

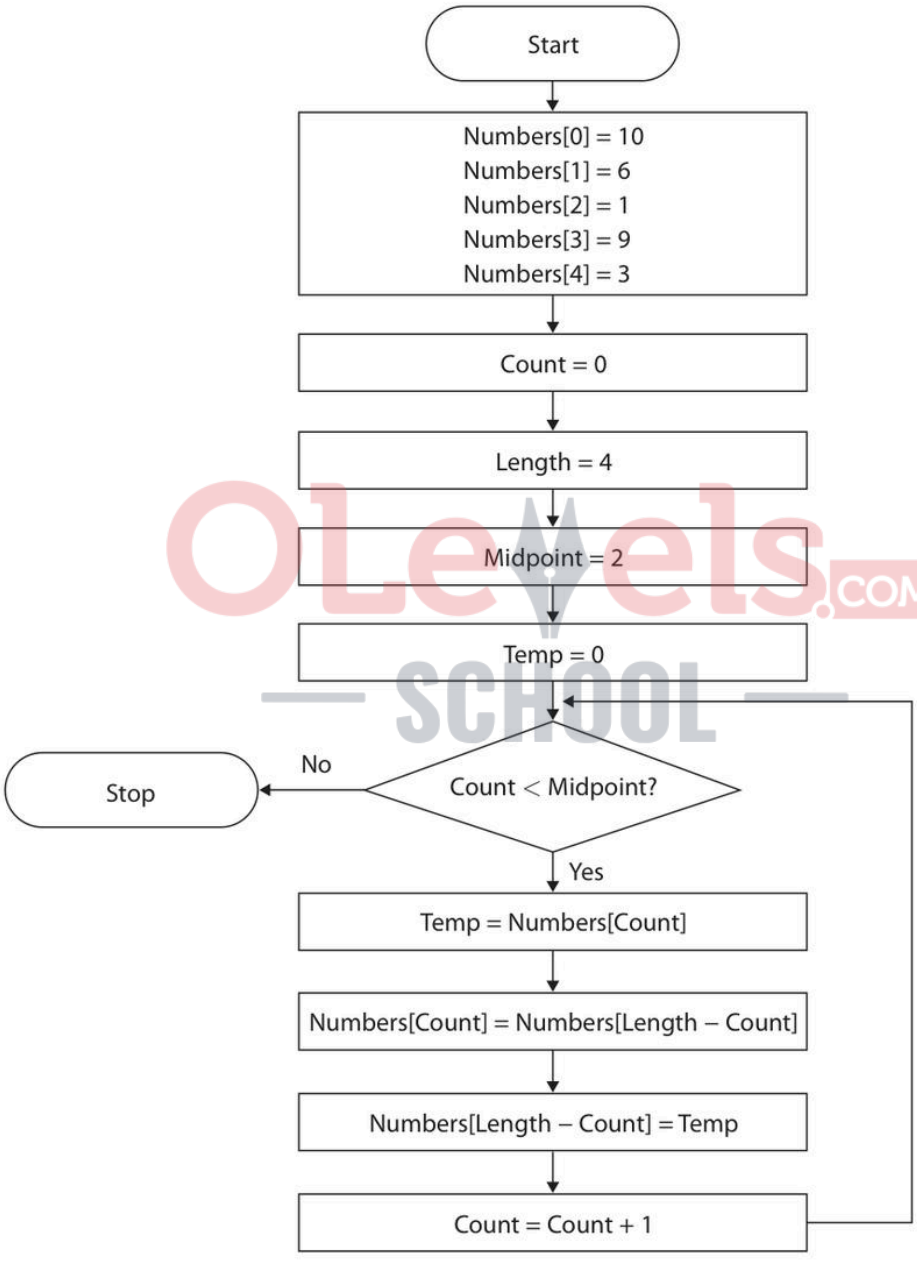


Figure 3

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 2 2 0

5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

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14

P 6 6 4 8 9 A 0 1 4 2 0

5. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

4 Programs can be used to manipulate data.

(a) A program is needed to convert an 8-bit binary number into denary.

The program must:

- receive the input of a binary pattern and validate that it has exactly eight characters (there is no need to validate that it only contains 0s and 1s)
- request the input again if the binary pattern does not have exactly eight characters
- when a valid binary pattern is input:
  - convert the binary pattern into its denary equivalent
  - print a message that contains the binary pattern and its denary equivalent
- include a comment to explain how the conversion process works.

HINT

Denary placeholder	128	64	32	16	8	4	2	1
Binary number	1	0	1	1	0	1	0	0
Denary number	128	0	32	16	0	4	0	0

128 + 32 + 16 + 4 = 180

Open **Q04a** in the code editor.

Write a program to implement these requirements.

You must use the structure given in **Q04a** to write the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(11)

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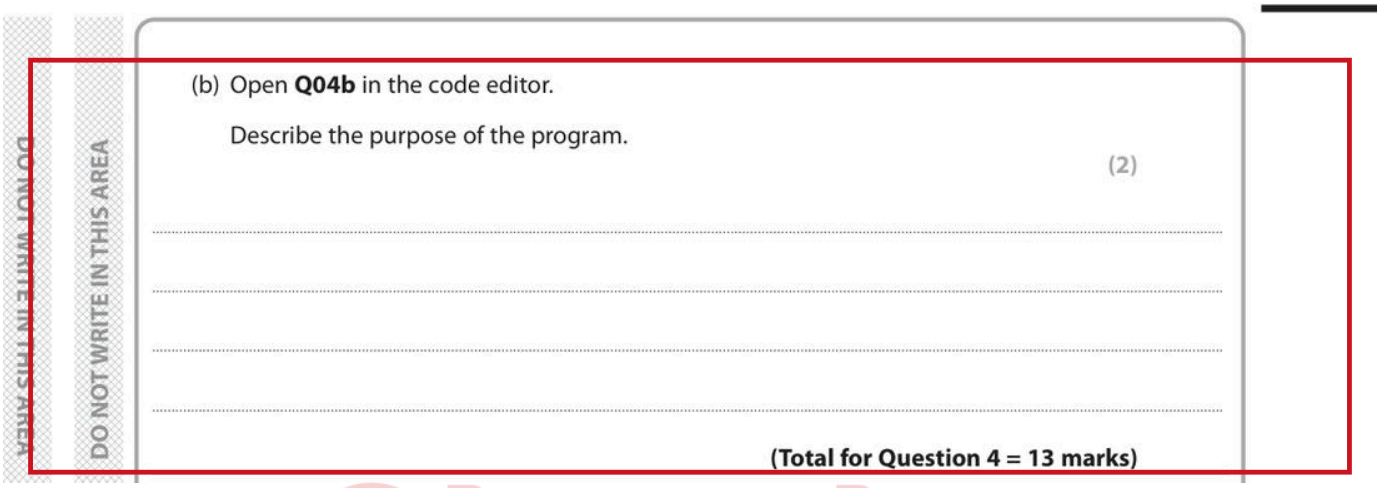
10

P 6 9 2 9 8 A 0 1 0 1 6

6. Question 4 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.



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(b) Open **Q04b** in the code editor.

Describe the purpose of the program.

(2)

(Total for Question 4 = 13 marks)

Working and answer

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7. Question 6 · part b · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

.....

Justification

.....

.....

.....

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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20

P 7 2 9 3 7 A 0 2 0 2 4

7. Question 6 · part b · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b only.

(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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7. Question 6 · part b · 1 mark · Working and answer

Working and answer

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8. Question 6 · part c.i · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.i only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8. Question 6 · part c.i · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

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8. Question 6 · part c.i · 2 marks · Working and answer

Working and answer

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9. Question 6 · part c.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 6 · part c.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

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(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

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(ii) State the purpose of the algorithm in **Figure 2**.

(1)

P 7 2 9 3 7 A 0 2 1 2 4

21

Turn over ▶

9. Question 6 · part c.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

(d) Cloud storage has become an increasingly popular service.

Discuss the benefits and drawbacks of using cloud storage.

(6)

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(Total for Question 6 = 12 marks)

9. Question 6 · part c.ii · 1 mark · Working and answer

Working and answer

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10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

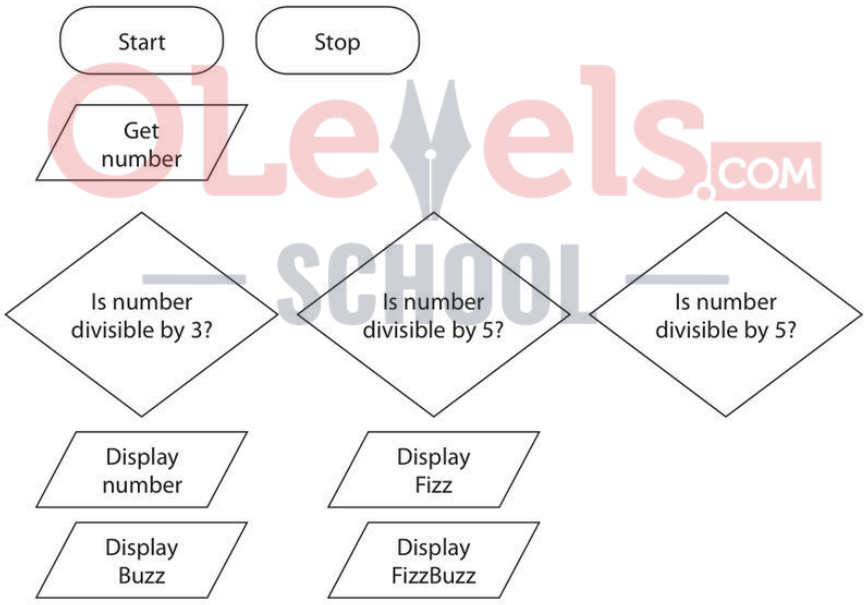
(d) In the game FizzBuzz a user enters a number.

These conditions apply:

- if the number is divisible by 3 the program displays Fizz
- if the number is divisible by 5 the program displays Buzz
- if the number is divisible by both 3 and 5 the program displays FizzBuzz
- if the number is not divisible by either 3 or 5, the program displays the number entered.

Hint: If one number is divisible by another number there is no remainder.

These are the components needed for the flowchart.



Draw the flowchart for the algorithm used in the game in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 5 7 3 6 A 0 2 4 2 8

10. Question 6 · part c · 1 mark · Working and answer

Working and answer

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11. Question 6 · part b.iii · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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11. Question 6 · part b.iii · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

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(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

.....

.....

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

.....

.....

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.....

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(Total for Question 6 = 13 marks)

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11. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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